

Part 8 Debt Financing

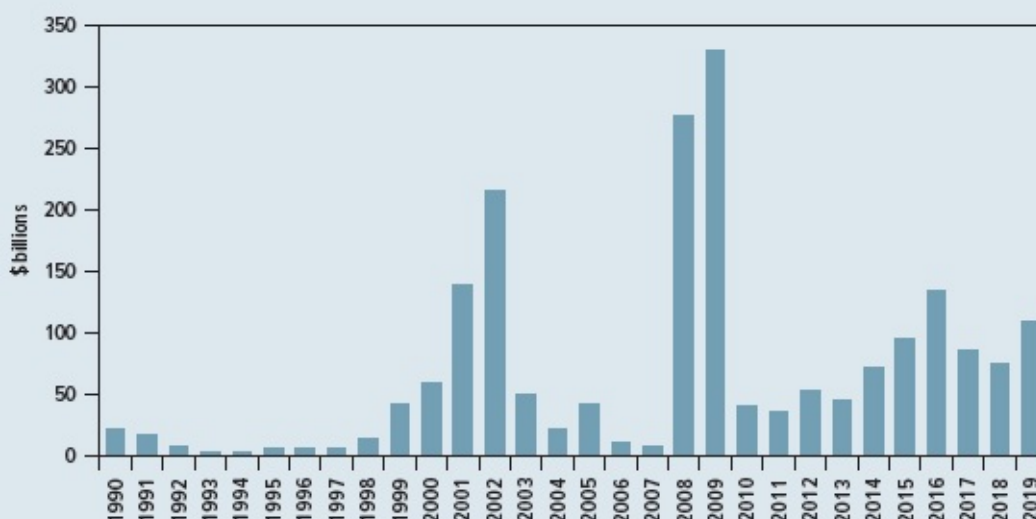
CHAPTER

24

Credit Risk and the Value of Corporate Debt

We first looked at how to value bonds way back in Chapter 3. We explained in that chapter what bond dealers mean when they refer to spot rates of interest and yields to maturity. We discussed why long-term and short-term bonds may offer different rates of interest and why prices of long-term bonds are affected more by a change in rates. We looked at the difference between nominal and real (inflation-adjusted) interest rates, and we saw how interest rates respond to changes in the prospects for inflation.

All the lessons of Chapter 3 hold good for both government and corporate bonds, but there is also a fundamental distinction between government and corporate issues. When a government borrows money, you can usually be confident that the debt will be repaid in full and on time. This is not true of corporate borrowing. Look, for example, at Figure 24.1. You can see that in 2009, following the financial crisis, companies defaulted on a record \$330 billion of debt. Bondholders are aware of the danger that they will not get their money back and so demand a higher yield. We turn now to consider how much higher these yields are and what determines their level.



► FIGURE 24.1

Global face value of defaulting debt, 1990–2020, (\$ billions).

Source: Moody's Investor Service, "Annual Default Study," February 2021.

Section 24-1 Yields on corporate debt

In Section 24-1, we begin our review of corporate bonds by looking at how yields vary with the likelihood of default. We show that you can think of the yield spread over government bonds as the amount that would be needed to insure against default. Lenders sometimes use credit default swaps to provide this insurance.

Section 24-2 Valuing the option to default

In Section 24-2, we look more carefully at the company's decision to default. We show that default is an *option*; if the going becomes too tough, the company has the option to stop payments on its bonds and hand over the business to the debtholders. We know what determines the value of options; therefore, we know the basic variables that must enter into the valuation of corporate bonds.

Section 24-3 Predicting the probability of default

In Section 24-3, we look at bond ratings and we describe some statistical and structural models that are used to estimate the probability that a corporate bond will default.



24-1 Yields on Corporate Debt

Following its introduction as a public company in 2014, California Resources Corporation incurred large losses and built up a heavy debt load. By 2019, its 8.0% bonds maturing in 2022 were trading at 24% of face value, where they offered a yield to maturity of 54%. A naïve investor who compared this figure with the 2% yield on Treasury bonds might have concluded that the bonds were a wonderful investment. But the owner would earn a 54% return only if the company repaid the debt in full. By 2019, that was looking increasingly doubtful. Because there was a significant risk that the company would default on its bonds, the *expected* yield was much less than 54%.

Distinguishing Promised and Expected Yields

Corporate bonds, such as the California Resources bond, offer a higher **promised yield** than government bonds, but do they necessarily offer a higher **expected yield**? We can answer this question with a simple numerical example. Suppose that the interest rate on one-year *risk-free* bonds is 5%. Backwoods Chemical Company has issued 5% notes with a face value of

\$1,000, maturing in one year. What will the Backwoods notes sell for?

If the notes are risk-free, the answer is easy—just discount principal (\$1,000) and interest (\$50) at 5%:

$$\text{PV of notes} = \frac{\$1,000 + 50}{1.05} = \$1,000$$

Suppose, however, that there is a 20% chance that Backwoods will default and that, if default does occur, holders of its notes receive half the face value of the notes, or \$500. In this case, the possible payoffs to the noteholders are

	Payoff	Probability
No default	\$1,050	0.8
Default	500	0.2

The expected payment is $0.8(\$1,050) + 0.2(\$500) = \$940$.

We can value the Backwoods notes like any other risky asset, by discounting their expected payoff (\$940) at the appropriate opportunity cost of capital. We might discount at the risk-free interest rate (5%) if Backwoods' possible default is totally unrelated to other events in the economy. In this case, default risk is wholly diversifiable, and the beta of the notes is zero. The notes would sell for

$$\text{PV of notes} = \frac{\$940}{1.05} = \$895$$

An investor who purchased the notes for \$895 would receive a *promised* yield of 17.3%:

$$\text{Promised yield} = \frac{\$1,050}{\$895} - 1 = 0.173$$

That is, an investor who purchased the notes for \$895 would earn a return of 17.3% if Backwoods does not default. Bond traders therefore might say that the Backwoods notes “yield 17.3%.” But the smart investor would realize that the notes' *expected* yield is only 5%, the same as on risk-free bonds.

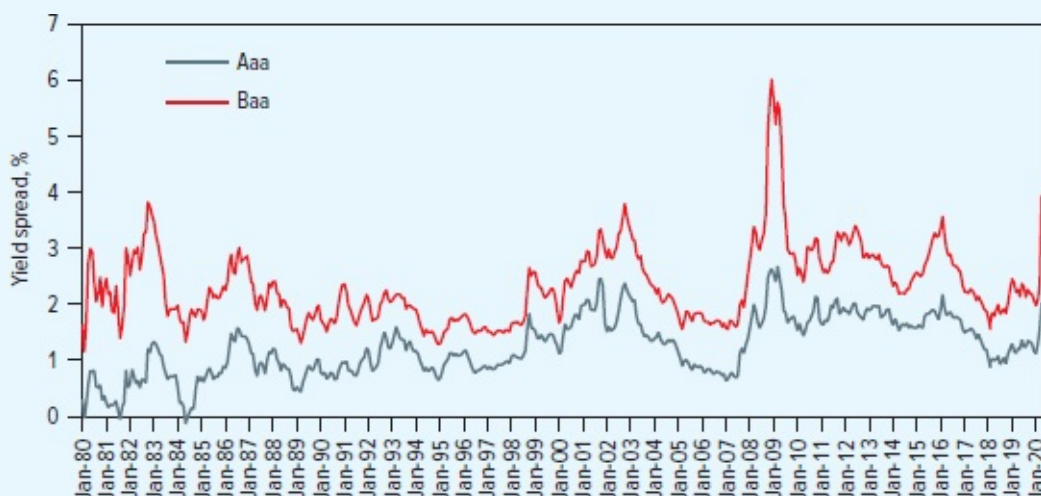
This, of course, assumes that the risk of default with these notes is wholly diversifiable so that they have no market risk. In general, risky bonds do have market risk (i.e., positive betas) because default is more likely to occur in recessions when all businesses are doing poorly. Suppose that investors demand a 3% risk premium giving an 8% expected rate of return. Then the Backwoods notes will sell for $940/1.08 = \$870$ and offer a promised yield of $(1,050/870) - 1 = 0.207$, or 20.7%.

24.1 Self-Test

Suppose that investors receive nothing if Backwoods defaults. If bondholders still demand an 8% expected return, what is the price of the bond, and what is its promised yield?

What Determines the Yield Spread?

Figure 24.2 shows how the yield spread on U.S. corporate bonds varies with the bond's risk. Bonds rated Aaa by Moody's are the highest grade bonds and are issued only by a few blue-chip companies. The promised yield on these bonds has on average been about 1% higher than the yield on Treasuries. Baa bonds are rated three notches lower; the yield spread on these bonds has averaged a little more than 2%. At the bottom of the heap are high-yield or "junk" bonds. There is considerable variation in the yield spreads on junk bonds; a typical spread might be about 6% over Treasuries, but spreads can rocket skyward as companies fall into distress.



► **FIGURE 24.2**

Monthly yield spreads between corporate and 10-year Treasury bonds, 1980–2020.

Source: The Federal Reserve Bank of St. Louis, <https://fred.stlouisfed.org/>.

Remember these are promised yields; companies don't always keep their promises and many high-yielding bonds have defaulted.

Figure 24.2 also shows that yield spreads can vary quite sharply from one year to the next, particularly for low-rated bonds. For example, they were unusually high in 2000–2002 and 2008–2009. Why is this? The main reason is that, as Figure 24.1 shows, these were periods when defaults were more likely. However, the fluctuations in spreads appear to be too large to be due simply to changing probabilities of default. It seems that there are occasions when investors are particularly reluctant to bear the risk of low-grade bonds and so scurry to the safe haven of government debt.¹

To understand more precisely what the yield spread measures, compare these two strategies:

Strategy 1: Invest \$1,000 in a floating-rate default-free bond yielding 9%.²

Strategy 2: Invest \$1,000 in a floating-rate corporate bond yielding 10%. At the same time, take out an insurance policy to protect yourself against the possibility of default. You pay an insurance premium of 1% a year, but in the event of default, you are compensated for any loss in the bond's value.

The two strategies offer similar payoffs. In the case of strategy 2, you gain a 1% higher yield, but this is exactly offset by the 1% annual premium on the insurance policy. Why does the insurance premium have to be equal to the spread? Because, if it was not equal, one strategy would dominate the other and there would be an arbitrage opportunity. The law of one price tells us that two equivalent risk-free investments must cost the same.

Our example tells us how to interpret the yield spread on corporate bonds. It is equal to the annual premium that would be needed to insure the bond against default.³

Insuring Bonds with a CDS It is possible to insure corporate bonds; you do so with an arrangement called a **credit default swap (CDS)**. If you buy a default swap, you commit to pay a regular insurance premium (or *spread*). For example, in January 2020 the spread on a Boeing CDS was quoted at 45 **basis points**. This meant that you needed to pay \$45,000 a year to buy protection on \$10 million worth of the company's debt ($\$45,000 = 0.0045 \times \10 million). In return, if Boeing subsequently defaulted on its debt, the seller of the swap would pay you the difference between the face value of the debt and its market value. You gain a higher yield by investing in Boeing bonds rather than Treasuries, but that gain is eaten up by the cost of insuring against default. In practice, for a variety of reasons, the CDS spread generally differs somewhat from the yield spread on the bond. This difference is termed the *basis*. For example, if the CDS price is less than the bond spread, the basis is said to be negative.

24.2 Self-Test

You are worried about the risk of default on your bond investments. Do you buy or sell credit default swaps?

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What Exactly Is a Default?

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The AIG CDS Debacle Credit default swaps offer a potentially valuable way for investors to

isolate and transfer credit risk. They, therefore, proved very popular, particularly with banks that needed to reduce the risk of their loan books. From almost nothing in 2000, the notional value of default swaps and related products mushroomed to \$62 trillion by the start of the financial crisis.⁵ Many of these default swaps were sold by **monoline insurers**, Page 695 which specialize in providing services to the capital markets. The monolines had traditionally concentrated on insuring relatively safe municipal debt but had been increasingly prepared to underwrite corporate debt, as well as many securities that were backed by subprime mortgages. By 2008, insurance companies had sold protection on \$2.4 trillion of bonds. As the outlook for many of these bonds deteriorated, investors began to question whether the insurance companies had sufficient capital to make good on their guarantees.

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A Most Controversial Trade

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FINANCE IN PRACTICE

Dynegy's Credit Default Swaps

› In the event of default, the seller of a credit default swap pays the buyer the difference between the face value of the debt and its market value. But finding the market value of defaulting bonds can be tricky. The following example illustrates how this is done.

In 2011, the electric utility, Dynegy, filed for bankruptcy with more than \$2 billion of debt and over \$700 million net of credit default swaps outstanding. The market value of Dynegy's debt and therefore the loss suffered by the CDS sellers was determined by a two-stage auction. In the first stage, all the participants submitted prices at which they would be prepared to sell or buy a minimum amount of a pre-approved pool of Dynegy bonds. As it turned out, most participants in this auction wished to buy rather than sell the bonds. So a second-stage auction was held to find a price that would tempt sufficient additional sellers.

The resulting equilibrium price from this second stage was 71.25% of face value, so the implied loss on each bond was $100 - 71.25 = 28.75\%$ of face value. Therefore, an investor who had bought a **notional value** of \$10 million of Dynegy credit default swaps was entitled to receive compensation from the seller of \$2.875 million.⁴

One of the largest providers of credit protection was AIG Financial Products, part of the

giant insurance group, AIG, with a portfolio of more than \$440 billion of credit guarantees. AIG’s clients never dreamt that the company would be unable to pay up: Not only was AIG triple-A rated, but it had promised to post generous collateral if the value of the insured securities dropped or if its own credit rating fell. So confident was AIG of its strategy that the head of its financial products group claimed that it was hard “to even see a scenario within any kind of realm of reason that would see us losing one dollar in any of these transactions.” But in September 2008, the unthinkable scenario occurred when the credit rating agencies downgraded AIG’s debt, and the company found itself obliged to provide \$32 billion of additional collateral within 15 days. Had AIG defaulted, everyone who had bought a CDS contract from the company would have suffered large losses on these contracts. To save AIG from imminent collapse, the Federal Reserve stepped in with an \$85 billion rescue package.

24-2

Valuing the Option to Default

The difference between a corporate bond and a Treasury bond is that the company has the option to default, whereas the government supposedly doesn’t.⁶ That default option is valuable. If you don’t believe us, think about whether (other things equal) you would prefer to be a shareholder in a company with limited liability or in a company with Page 696 unlimited liability. Of course, you would prefer to have the option to walk away from your company’s debts. Unfortunately, every silver lining has its cloud, and the drawback to having a default option is that corporate bondholders expect to be compensated for giving it to you. That is why corporate bonds sell at lower prices and offer higher yields than government bonds.

We can illustrate the nature of the default option by returning to the plight of Circular File Company, which we discussed in Chapter 17. Circular File borrowed \$50 per share, but then fell on hard times, and the market value of its assets fell to \$30. Circular’s bond and stock prices fell to \$27 and \$3, respectively. Thus, Circular’s *market-value* balance sheet is:

Circular File Company			
Asset value	\$30	\$27	Bond
		3	Stock
	\$30	\$30	Firm value

If Circular’s debt were due and payable now, the firm could not repay the \$50 it originally borrowed. It would default, leaving bondholders with assets worth \$30 and shareholders with nothing. The reason that Circular stock has a market value of \$3 is that the debt is *not* due immediately, but one year from now. A stroke of good fortune could increase firm value enough to pay off the bondholders in full, with something left over for the stockholders.

Circular File is not compelled to repay the debt at maturity. If the value of its assets is less

than the \$50 that it owes, it will choose to default on the debt and the bondholders will take over the company's assets. Thus, the company has, in effect, issued a safe bond but, at the same time, acquired an option to sell the firm's assets to the bondholders for the amount of the debt. The exercise price of this put option is \$50, the face value of the bond. If the value of the company's assets when the bond matures is greater than \$50, Circular will not exercise its option to default. If the assets' value is less than \$50, it will pay Circular to exercise its option and to hand over the assets in settlement of the debt.

Bond traders, investors, and financial managers refer to *default puts*. When a firm defaults, its stockholders are, in effect, exercising their default put. The put's value is the value of limited liability—the value of the stockholders' right to walk away from their firm's debts in exchange for handing over the firm's assets to its creditors. In the case of Circular File, this option to default is extremely valuable because default is likely to occur. At the other extreme, the value of IBM's option to default is trivial compared with the value of IBM's assets. Default on IBM bonds is possible but extremely unlikely. Option traders would say that for Circular File, the put option is “deep in the money” because today's asset value (\$30) is well below the exercise price (\$50). For IBM, the put option is far “out of the money” because the value of IBM's assets greatly exceeds the amount of IBM's debt.

24.3 Self-Test

Who owns the default put option—the shareholders or the bondholders?

Finding Bond Values

Valuing corporate bonds should be a two-step process:

$$\text{Bond value} = \text{bond value assuming no chance of default} - \text{value of put option on assets} \quad (24.1)$$

The first step is easy: Calculate the bond's value assuming no default risk. (Discount promised interest and principal payments at the rate offered by Treasuries of the same maturity.) The second step requires you to calculate the value of a put written on the firm's assets, where the maturity of the put equals the maturity of the bond and the exercise price of the put equals the promised payment.

In Chapter 17, we *assumed* that the value of Circular's bond was \$27. Now we can see where that figure could have come from. Suppose that the standard deviation of the returns on Circular's assets is 60% a year and that the risk-free interest rate is 10%. The current value of Circular's assets is \$30, and the exercise price of the default option is \$50. If you enter these data into the Black–Scholes model, you find that the value of Circular's default

put is \$18.6. You can now value its bond:

$$\begin{aligned}\text{Value of Circular's bond} &= \text{value of default-free bond} - \text{value of default put} \\ &= 50/1.1 - 18.6 \\ &= \$26.9 \text{ (about \$27)}\end{aligned}$$

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Try It! Valuing the default put

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Before you get too excited about valuing the default option, we should warn you that, in practice, you would encounter complications that make the valuation of corporate bonds considerably more difficult than it sounds. For example, we assumed that Circular File is committed to making a single payment of \$50 at the end of the year. But suppose, instead, that it has issued a 10-year bond that pays interest annually. In this case, there are 10 payments rather than just 1. When each payment comes due, Circular has the option to make the coupon payment or to default. If it makes the first payment, Circular obtains a second option to default when the second interest payment becomes due. The reward to making this payment is that the stockholders get a third put option, and so on. (This is an example of a *compound* put option.)

Of course, if the firm does not make any of these payments when due, bondholders take over, and stockholders are left with nothing. In other words, if Circular decides to exercise its default option, it gives up all subsequent default options.

Valuing the 10-year bond when it is issued is equivalent to valuing the first of the 10 options. But you cannot value the first option without valuing the nine that follow. Even this example understates the practical difficulties because large firms may have dozens of outstanding debt issues with different interest rates and maturities, and before the current debt matures, they may make further issues. Consequently, when bond traders evaluate a corporate bond, they do not reach for their option calculator. They are more likely to start by identifying bonds with similar maturity and risk of default and look at the yield spreads offered by these bonds.

Valuing the default put may be challenging, but, now we know that limited liability is an option, we also know what the value of that option depends on. The following table shows how the value of the option to default depends on the underlying variables:⁷

If there is an increase in:	Value of default put:
Value of company's assets	Declines
Standard deviation of asset value	Rises
Amount of outstanding debt	Rises
Debt maturity	Rises

Default-free interest rate	Declines
Dividend payments	Rises

We have seen that corporate bonds sell at higher promised yields than Treasury bonds with the same maturities. Can we explain the size of these yield spreads in terms of the default put that attaches to corporate bonds? The answer appears to be “yes” in the case of investment-grade bonds. For these bonds, the default risk can do a fairly good job of explaining the typical yield spread.⁸ However, yields on junk bonds appear to be higher than their default experience would justify. It seems that investors in junk bonds demand additional yield to compensate for their relative lack of liquidity.

FINANCE IN PRACTICE

Valuing Government Financial Guarantees

› When American Airlines declared bankruptcy in 2011, its pension plan had liabilities of \$18.5 billion and assets of just \$8.3 billion. But the 130,000 workers and retirees did not face a destitute old age. Their pensions were largely guaranteed by the Pension Benefit Guaranty Corporation (PBGC). This is a U.S. federally chartered corporation created in 1974 to ensure the continuity of pension payments in the event of the bankruptcy of an employer.

Pension promises are a long-term liability just like the promises to bondholders. The guarantee by the PBGC changes the pension promises from a risky liability to a safe one. If the company goes belly-up and there are insufficient assets to cover the pensions, the PBGC makes up most of the difference.⁹

The government recognizes that the guarantee provided by the PBGC is costly. Thus, shortly after assuming the liability for the American Airlines plan, the PBGC calculated that the discounted value of payments on all defaulted plans and those close to default amounted to \$98 billion.

Unfortunately, these calculations ignore the risk that other firms in the future may fail and hand over their pension liability to the PBGC. To calculate the cost of the guarantee, we need to think about what the value of company pension promises would be without any guarantee:

$$\begin{aligned} \text{Value of guarantee} &= \text{value of guaranteed pensions} \\ &\quad - \text{value of pension promises} \\ &\quad \text{without a guarantee} \end{aligned}$$

With the guarantee, the pensions are as safe as a promise by the U.S. government¹⁰; without the guarantee, the pensions are like an ordinary debt obligation of the firm. We already know what the difference is between the value of safe government debt and risky corporate debt. It is the value of the firm's right to hand over the assets of the firm and to walk away from its obligations. Thus, the value of the pension guarantee is the value of this put option.

A paper prepared for the Congressional Budget Office showed how option pricing models can help to give a better measure of the cost to the PBGC of pension guarantees.¹¹ Their estimates suggest that the value of the PBGC's guarantees was much higher than its published estimate.

The PBGC is not the only government body to provide financial guarantees. Fortunately, option pricing is leading to a better way to calculate their cost.

The Value of Corporate Equity

We saw in Chapter 21 that the value of a put option is identical to the value of a call option with the same exercise price, plus the present value of the exercise price, and less the value of the underlying asset. Think what this means for the value of Circular File's default put:

$$\begin{aligned} \text{Value of put option to default} &= \text{value of call option on Circular's assets} \\ &\quad + \text{present value of Circular's debt at the risk-free interest rate} \\ &\quad - \text{value of Circular's assets} \end{aligned}$$

If you twist this formula around, you get:

$$\begin{aligned} \text{Value of call option} &= \text{value of} & - & \left(\text{present value of} & - & \text{value of put} \right) \\ \text{on Circular's assets} & \text{Circular's assets} & & \text{Circular's debt at the} & \text{option to default} \\ & & & \text{risk-free interest rate} & \\ & = \text{value of} & - & \text{value of} & = \text{value of Circular's} \\ & \text{Circular's assets} & & \text{Circular's debt} & \text{equity} \end{aligned}$$

The expression on the right-hand side is simply the value of Circular's equity. This tells us that you can think of Circular's bondholders as effectively owning the company now, but the shareholders have the option to buy it back from them at the end of the year by paying off the debt. Thus, the balance sheet of Circular File can be expressed as follows:

Circular File Company (Market Values)			
Asset value	\$30	\$26.9	Bond value = asset value — value of call
		3.1	Stock value = value of call
	\$30	\$30	Firm value = asset value

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Try It! Leverage and debt betas

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24.4 Self-Test

Describe the value of corporate debt and equity in terms of the value of the company's assets and a call option on these assets. What is the exercise price of the call? Assume that the company has just one issue of debt that is due to make just one future payment.

24-3 Predicting the Probability of Default

Banks and other financial institutions not only want to know the value of the loans that they have made, but they also need to know the risk that they are incurring. Some rely on the judgments of specialized bond-rating services. Others have developed their own models for measuring the probability that the borrower will default. We describe bond ratings first and then discuss two classes of models for predicting default.

As discussed in Chapter 3, the relative quality of most traded bonds can be judged by bond ratings. There are three principal rating services—Moody's, Standard & Poor's, and Fitch.¹² Table 24.1 summarizes these ratings. For example, the highest quality bonds are rated triple-A (Aaa) by Moody's, then come double-A (Aa) bonds, and so on. Bonds rated Baa or above are known as *investment-grade* bonds.¹³ Many pension funds, and other financial institutions are not allowed to invest in bonds unless they are investment-grade.¹⁴

Moody's	Standard & Poor's and Fitch
Investment-Grade Bonds	
Aaa	AAA

Aa	AA
A	A
Baa	BBB
Junk Bonds	
Ba	BB
B	B
Caa	CCC
Ca	CC
C	C

TABLE 24.1 Key to bond ratings. The highest quality bonds are rated triple-A. Investment-grade bonds have to be the equivalent of Baa or higher. Bonds that don't make this cut are called "high-yield" or "junk" bonds.

Bonds rated below Baa are termed **high-yield** or **junk bonds**. Most junk bonds used to be *fallen angels*—that is, bonds of companies that had fallen on hard times. But during the 1980s, new issues of junk bonds multiplied 10-fold, as more and more companies issued large quantities of low-grade debt to finance takeovers. The result was that for the first time, corporate midgets were able to raise the money to take control of corporate giants.

Issuers of these junk bonds often had debt ratios of 90–95%. Many worried that this threatened the health of corporate America, and as default rates on corporate debt rose to 10% in the early 1990s, the market for new issues of junk bonds dried up. Since then, the market for junk debt has had its ups and downs. For example, as interest rates on Treasuries dwindled in the years following the financial crisis, investors sought higher yields, and new issues of junk bonds boomed.

Bond ratings are judgments about firms' financial and business prospects. There is no fixed formula that is used to calculate ratings. Nevertheless, investment bankers, bond portfolio managers, and others who follow the bond market closely can get a fairly good idea of how a bond will be rated by looking at a few key numbers, such as the firm's operating margin, the return on assets, the debt ratio, and the ratio of earnings to interest. Table 24.2 shows how these ratios vary with the firm's bond rating.

Ratio	Aaa	Aa	A	Baa	Ba	B	Caa-C
Operating margin (%)	24.8	24.2	16.9	15.0	11.2	7.6	3.8
Pretax return on assets (%)	12.3	14.7	13.5	10.4	8.8	6.6	4.9
Long- plus short-term debt ratio (%)	42.8	41.5	46.7	48.4	55.9	68.1	94.4
Interest coverage	14.4	24.4	13.7	9.8	6.5	3.3	1.7

TABLE 24.2 How financial ratios differ according to a firm's bond rating. Median ratios for U.S. non-financial firms by bond rating.

Source: Moody's Financial Metrics, "Key Ratios by Rating and Industry for North American Non-Financial Corporations," December 2017.

24.5 Self-Test

Which of these statements describe(s) investment-grade bonds?

- They generally have lower interest coverage.
- They generally have lower debt ratios.
- They generally have higher operating margins.
- They generally have lower return on assets.

FINANCE IN PRACTICE

Country Credit Ratings

In addition to rating corporate debt, the major credit agencies also assign a rating for sovereign, state and municipal debt. Table 24.3 shows some examples of how Moody's rated the debt of different governments in September 2021.

Rating	Countries	
Aaa	Germany	USA
Aa	France	UK
A	China	Japan

Baa	India	Italy
Ba	Brazil	South Africa
B	Turkey	Pakistan
Caa	Cuba	El Salvador

TABLE 24.3 Examples of Moody's ratings for foreign currency government debt, September 2021.

Source: www.moody.com.

Countries with highly rated debt share some common characteristics. They have high per capita income, low inflation and low external debt, a high level of economic development, and no history of past defaults.¹⁵

Figure 24.3 shows that bond ratings do reflect the probability of default. Since 1970, no U.S. bonds that were initially rated triple-A by Moody's have defaulted in the year after issue, and only 1 in 700 have defaulted within 10 years of issue. (The Aaa default rate is not plotted in Figure 24.3. It would be invisible.) At the other extreme, about half of Caa to C bonds have defaulted by year 10. Of course, bonds do not usually fall suddenly from grace. As time passes and the company becomes progressively more shaky, the agencies revise downward the bond's rating to reflect the increasing probability of default.

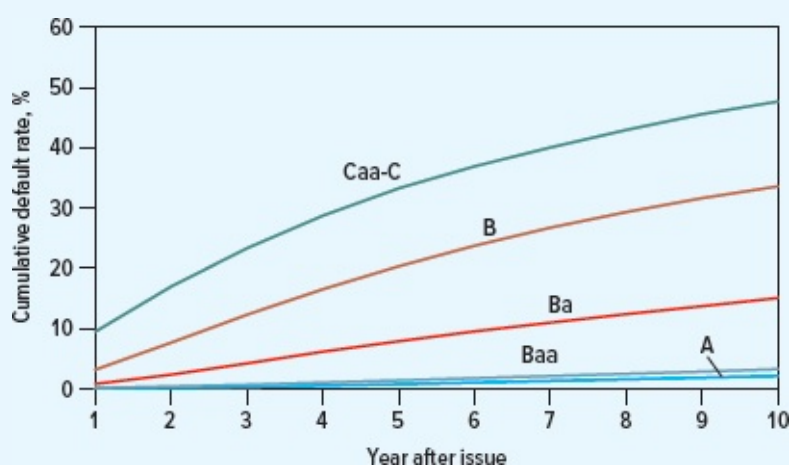


FIGURE 24.3

Global default rates of corporate bonds 1983–2020, by Moody's rating at time of issue.

Source: Moody's Investor Service, "Annual Default Study: 1920–2020," February 2021.

Rating agencies don't always get it right. When Enron went belly-up in 2001, investors

protested that only two months earlier the company's debt had an investment-grade rating. Rating agencies also did not win many friends during the financial crisis of 2007–2009, when many of the mortgage-backed securities that had been given triple-A ratings defaulted. Conversely, when agencies *do* downgrade a company's debt, they are often accused of overhasty action that increases the cost of borrowing.

Statistical Models of Default

Financial managers pay attention to bond ratings, but they may also make their own judgments about the chances of default. There are two methods that they use to do this. We first describe statistical models to predict default, and we then describe structural models.

If you apply for a credit card or a bank loan, you will probably be asked to complete a questionnaire that provides details about your job, home, and financial health. This information is then used to calculate an overall credit score.¹⁶ If you do not make the grade on the score, you are likely to be refused credit or subjected to a more detailed analysis. In a similar way, mechanical credit scoring systems are used by banks to assess the risk of their corporate loans and by firms when they extend credit to customers.

Suppose that you are given the task of developing a system that will help to decide which businesses are poor credits. You start by examining the financial statements of companies that went bankrupt over the past 20 years. Figure 24.4 shows what you find. You can see that, on average, bankruptcy was preceded by declining profit margins and return on capital, increasing financial leverage, and deteriorating liquidity.

Rather than focusing on individual ratios, it makes sense to combine the ratios into a single score that can separate the creditworthy sheep from the impecunious goats. In other words, you need to estimate an equation that relates the risk of bankruptcy to a set of different financial variables. Most statistical bankruptcy models focus on a relatively small set of accounting ratios, such as measures of leverage, profitability, and liquidity. For small businesses, there may be little alternative to the use of accounting data, but for large, publicly traded firms, it is also possible to take advantage of the information in security prices. A declining share price, volatile returns, a low market-to-book ratio, and a low share price all seem to indicate a higher probability of impending bankruptcy.¹⁷

One of the earliest statistical models that is still widely used is the Z-score model proposed by Edward Altman.¹⁸ A company's Z-score is given by the following formula:

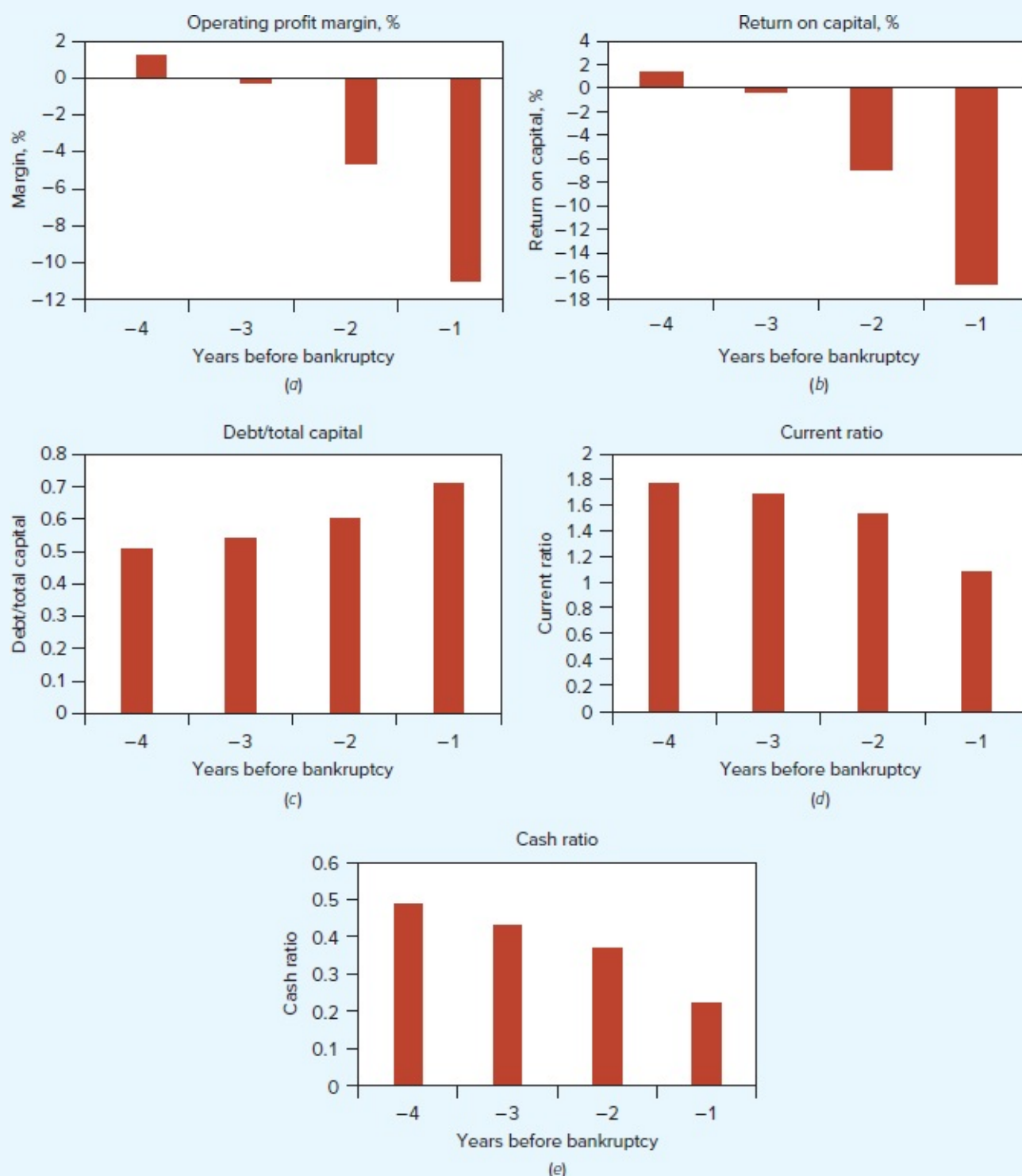
$$Z = 1.2x_1 + 1.4x_2 + 3.3x_3 + 0.6x_4 + 1.0x_5$$

where:

- x_1 = working capital/total assets
- x_2 = retained earnings/total assets
- x_3 = earnings before interest and tax (EBIT)/total assets
- x_4 = market value of equity/total liabilities
- x_5 = sales/total assets

Altman suggested that a score below 1.8 indicated impending bankruptcy.

Statistical models of bankruptcy use a variety of techniques to estimate the relationship between the occurrence of bankruptcy and the set of financial variables. Altman's model used multiple discriminant analysis to calculate the Z-score. Others have used more sophisticated hazard or probit models.



► FIGURE 24.4

Median financial ratios of 409 firms filing for bankruptcy between 1999 and 2019.

Source: Compustat and WRDS Financial Ratios.

Before we leave the topic of these statistical models, we should issue a health warning. When you construct a risk index, it is tempting to experiment with many different combinations of variables until you find the equation that would have worked best in the past. Unfortunately, if you “mine” the data in this way, some of the relationships that you uncover are likely to be due to chance, and therefore the system will work less well in the future. If you are misled by past successes into placing too much faith in your model, you could be worse off than if you had pretended that you could not tell one would-be borrower from another and extended credit to all of them. Does this mean that firms should not use credit scoring systems? Not a bit. It merely implies that it is not sufficient to have a good system; you also need to know how much to rely on it.

Structural Models of Default

The common feature of statistical default models is that the user picks a number of variables that she suspects might indicate approaching financial distress and then uses statistical methods to find the combination of variables that best predicts which firms will become bankrupt.

A different approach is to develop a structural model that builds on the insight that shareholders will exercise their option to default if the market value of the assets falls below the payments that must be made on the debt. The best known of these models is the Merton model, named after Robert Merton who first developed it.¹⁹ It is also called the Moody’s KMV model after the firm that produced a commercial version. We will illustrate with a simple example.

Imagine a company, call it Upsilon, whose assets have a current market value of \$100. Its debt has zero coupon, a face value of \$60, and matures in one year. The return on the assets has a standard deviation of 30%, so the asset value when the debt matures could be more or less than the face amount of the debt. The risk-free rate of interest is 5%.

If the debt was risk-free, it would be worth $60/1.05 = \$57.14$. But Upsilon’s debt is risky: If the assets are worth less than \$60, the shareholders will exercise their option to default and hand over the assets to the debtholders. The Black–Scholes model tells us that the value of this put option is \$0.27. Therefore, the value of the debt is:

$$\begin{aligned}
 \text{Value of debt} &= \text{value of risk-free debt} - \text{value of default put} \\
 &= 57.14 - 0.27 \\
 &= \$56.87
 \end{aligned}$$

The value of the equity is therefore:

$$\text{Value of equity} = \text{Value of assets} - \text{value of debt} = 100 - 56.87 = 43.13$$

To estimate the *probability* of default, we need to calculate the probability that the company will choose to exercise its default put option. Figure 24.5 shows the distribution of possible asset values at the end of the year. It assumes that asset values are lognormally distributed with a standard deviation of 30%. It also assumes that investors are risk-neutral, so that the expected value of Upsilon's assets is $100 \times 1.05 = \$105$.²⁰

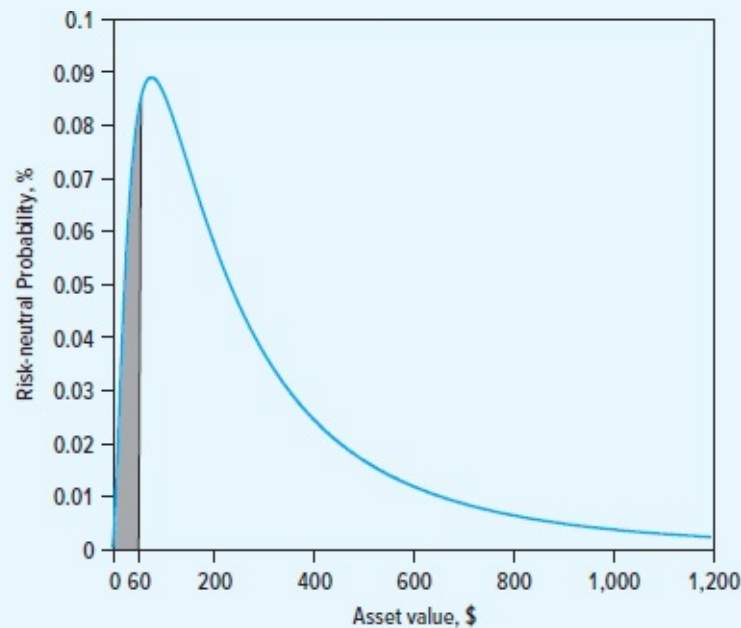


FIGURE 24.5

Upsilon has issued one-year debt with a face value of \$60. The shaded area shows that there is a 4.3% risk-neutral probability that the value of the company's assets at the end of the year will be less than \$60. In that case, the company will choose to default.

The shaded area shows the probability in a risk-neutral world that the value of the assets at the end of the year will be less than the face value of the debt and that Upsilon will therefore default. If you look back at the Black–Scholes formula in Section 22.3, you will see the expression $N(d_2)$. The risk-neutral probability that the option will be exercised is equal to $1 - N(d_2)$. In the case of Upsilon,

$$\text{Risk – neutral probability of default} = 1 - N(d_2) = 0.043, \text{ or } 4.3\% \quad (24.2)$$

There is a 4.3% risk-neutral chance that Upsilon will default.

24.6 Self-Test

Suppose that Upsilon's debt has a face value of \$70. Use the Black–Scholes app from Chapter 22 to recalculate the risk-neutral probability that it will default.

Structural models of default have obvious attractions. They have a theoretical base. So the relevant variables are pretty well known, and you do not need to go prospecting among past data to find variables that may be indications of impending default. But when you apply the model in practice, you inevitably encounter complications. For example, unless you can observe the market value of the company's debts, you can't observe the value of its assets or measure their volatility.²¹ Also, companies may have several debt issues, each with a different maturity. Nevertheless, it is possible to adapt the structural model to estimate the probability of default by real companies.²²



KEY TAKEAWAYS

- **Expected yield on a corporate bond.** Corporations have limited liability. Therefore, if companies are unable to pay their debts, they can file for bankruptcy. As a result lenders may receive less than they are owed and the *expected* yield on a corporate bond is less than the *promised* yield.
- **The yield spread on a corporate bond.** Because of the possibility of default, the promised yield on a corporate bond is higher than on a government bond. You can think of this yield spread as the amount that you would need to pay to insure the bond against default. There is a market for insurance policies that protect the debtholder against default. These policies are called credit default swaps. There are few free lunches in financial markets. So the extra yield you get for buying a corporate bond is likely to be eaten up by the cost of insuring against default.
- **The default put.** The company's option to default is equivalent to a put option. If the value of the firm's assets is less than the amount of the debt, it will pay the company to default and to allow the lenders to take over the assets in settlement of the debt. Thus

$$\text{Bond value} = \text{value of risk-free bond} - \text{value of put option on issuer's assets}$$

This insight tells us what we need to think about when valuing corporate debt—the

current value of the firm relative to the point at which it would default, the volatility of the assets, the maturity of the debt payments, and the risk-free interest rate. Unfortunately, most companies have several loans outstanding with payments due at different times. This considerably complicates the task of valuing the put option.

- **Bond ratings.** When investors want a measure of the risk of a company's bonds, they usually look first at the rating that has been assigned by Moody's, Standard & Poor's, or Fitch. They know that bonds with investment-grade ratings (at least Baa or BBB) are much less likely to default than bonds with a junk rating.
- **Statistical models of default.** Banks, rating services, and consulting firms have also developed models for estimating the likelihood of default. Statistical models take accounting ratios or other indicators of corporate health, and weight them to produce a single measure of default probability.
- **Structural default models.** Structural models, such as the Merton model, take a different tack and seek to measure the probability that the market value of the firm's assets will fall to the point at which the firm will choose to exercise its option to default rather than keep up with its debt payments.



FURTHER READING

The websites of the main credit rating agencies contain a variety of useful reports on credit risk. (See in particular www.moodys.com, www.standardandpoors.com, and www.fitchratings.com.)

Altman and Hotchkiss provide a review of credit scoring models in:

E. I. Altman, E. Hotchkiss, and W. Wang, *Corporate Financial Distress, Restructuring and Bankruptcy*, 4th ed. (New York: John Wiley, 2019).

Books that discuss corporate bonds and credit risk include:

A. Saunders and L. Allen, *Credit Risk Measurement*, 3rd ed. (New York: John Wiley, 2010).

J. B. Caouette, E. I. Altman, P. Narayanan, and R. Nimmo, *Managing Credit Risk*, 2nd ed. (New York: John Wiley, 2008).

D. Duffie, *Measuring Corporate Default Risk* (Oxford, UK: Oxford University Press, 2011).

D. Duffie and K. J. Singleton, *Credit Risk: Pricing, Measurement and Management* (Princeton, NJ: Princeton University Press, 2003).



PROBLEM SETS



connect

Select problems are available in McGraw Hill's *Connect*.

Please see the preface for more information.

1. **Expected yield (S24.1)** You own a 5% bond maturing in two years and priced at 87%. Suppose that there is a 10% chance that at maturity the bond will default and you will receive only 40% of the promised payment. What is the bond's promised yield to maturity? What is its expected yield (i.e., the possible yields weighted by their probabilities)?
2. **Bond ratings (S24.1)** In February 2018, Aaa bonds yielded 2.36%, Baa bonds yielded 3.43%, and Treasuries with the same maturities yielded 0.676%.
 - a. What was the credit spread on Aaa bonds?
 - b. What was the spread on Baa bonds?
 - c. What do you think would be the difference in price (as a percent of face value) between a typical 5% 10-year Baa bond and a similar Treasury bond?
3. **Bond ratings (S24.1)** It is 2030 and the yields on corporate bonds are as follows:

Aaa	A	Ba
8%	10%	12%

Tau Corp wishes to raise \$10 million by an issue of 9% 10-year bonds. What will be the likely issue price (as a percent of face value) if Tau is rated (a) Aaa, (b) A, or (c) Ba?

4. **Default option (S24.2)** The difference between the value of a government bond and a similar corporate bond is equal to the value of an option. What is this option, and what is its exercise price?
5. **Default option (S24.2)** Other things equal, would you expect the difference between the price of a Treasury bond and a corporate bond to increase or decrease with
 - a. The company's business risk?
 - b. The degree of financial leverage?
 - c. The time to maturity?
6. **Default option (S24.2)** Company A has issued a single zero-coupon bond maturing in 10 years. Company B has issued a coupon bond maturing in 10 years. Explain why it is more complicated to value B's debt than A's.
7. **Default option (S24.2)** How much would it cost you to insure the bonds of Backwoods Chemical against default? (See Section 24-1.)
8. **Default option (S24.2)** Digital Organics has 10 million outstanding shares trading at

\$25 per share. It also has a large amount of debt outstanding, all coming due in one year. The debt pays interest at 8%. It has a face value of \$350 million but is trading at a market value of only \$280 million. The one-year risk-free interest rate is 6%.

- a. Write out the put–call parity formula for Digital Organics’ stock, debt, and assets.
 - b. What is the value of the company’s option to default on its debt?
9. **Default option (S24.2)** Square File’s assets are worth \$100. It has \$80 of zero-coupon debt outstanding that is due to be repaid at the end of two years. The risk-free interest rate is 5%, and the standard deviation of the returns on Square File’s assets is 40% per year. Calculate the present value of the company’s debt and equity.
10. **Predicting default probability (S24.3)** A friend has mentioned that she has read somewhere that the following variables can be used to predict bankruptcy: (a) the company debt ratio; (b) the interest coverage; (c) the amount of cash Page 708 relative to sales or assets; (d) the return on assets; (e) the market-to-book ratio; (f) the recent return on the stock; (g) the volatility of the stock returns. The problem is that she can’t remember whether a high value of each variable implies a high or a low probability of bankruptcy. Can you help her out?

BEYOND THE PAGE



Try It! The Black–Scholes model

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11. **Predicting default probability (S24.3)** What variables are required to use the Merton model to calculate the risk-neutral probability that a company will default on its debt?
12. **Predicting default probability (S24.3)** Company X has borrowed \$150 maturing this year and \$50 maturing in 10 years. Company Y has borrowed \$200 maturing in five years. In both cases, asset value is \$140. Sketch a scenario in which X does not default but Y does.
13. **Predicting default probability (S24.3)** Discuss the problems with developing a numerical credit scoring system for evaluating personal loans. You can only test your system using data for applicants who have in the past been granted credit. Is this a potential problem?
14. **Predicting default probability (S24.3)** Look back at Section 24.3. Suppose that the standard deviation of the return on Upsilon’s assets is 50%. Recalculate the probability that the company will default.

BEYOND THE PAGE



Try It! The Black–Scholes model

CHALLENGE

15. **Default option (S24.2)** Look back at the first Backwoods Chemical example at the start of Section 24-1. Suppose that the firm's book balance sheet is

Backwoods Chemical Company (Book Values)			
Net working capital	\$ 400	\$1,000	Debt
Net fixed assets	<u>1,600</u>	<u>1,000</u>	Equity (net worth)
Total assets	\$2,000	\$2,000	Total value

The debt has a one-year maturity and a promised interest payment of 9%. Thus, the promised payment to Backwoods's creditors is \$1,090. The market value of the assets is \$1,200, and the standard deviation of asset value is 45% per year. The risk-free interest rate is 9%. Calculate the value of Backwoods's debt and equity.

BEYOND THE PAGE



Try It! The Black-Scholes model

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SOLUTIONS TO SELF-TEST QUESTIONS

- 24.1** The expected payoff is $0.8 \times \$1,050 = \840 . The bond price is $\$840/1.08 = \778 , and the promised yield is $1,050/778 - 1 = 0.35$, or 35%.
- 24.4** You *buy* a credit default swap.
- 24.3** The shareholders have the option to hand over the assets to the bondholders rather than to repay the bonds.
- 24.4** The value of the equity is a call option on the assets (i.e., shareholders get to own the assets by repaying the debt). The value of the debt is equal to the value of the assets less the value of the shareholders' call option. The exercise price of the call option is the face value of the company's debt.
- 24.5** b and c.
- 24.6** If the exercise price of the put is \$70, $N(d_2) = 0.8852$. So the risk-neutral probability of default is $1 - 0.8852 = 0.1148$, or 11.48%.



FINANCE ON THE WEB

1. Go to **finance.yahoo.com** and select three industrial companies that have been experiencing difficult times.
 - a. Are the companies' troubles reflected in their financial ratios? (You may find it helpful to refer to Figure 24.4.)
 - b. Now look at the company's bond rating. Do the ratings and ratios provide consistent messages?

¹For evidence on the effect of changing risk aversion on bond spreads, see A. Berndt, R. Douglas, D. Duffie, and M. Ferguson, "Corporate Credit Risk Premia," *Review of Finance* 22 (March 2018), pp. 419–454.

²The interest payment on floating-rate bonds goes up and down as the general level of interest rates changes. Thus, a floating-rate, default-free bond will sell at close to face value on each coupon date. Many governments issue "floaters." The U.S. Treasury does not do so, though some U.S. government agencies do.

³For illustration, we have used the example of a floating-rate bond to demonstrate the equivalence between the yield spread and the cost of default insurance. The yield spread on a fixed-rate bond should be very close to the cost of insurance (i.e. the CDS spread).

⁴For a detailed description of the Dynegy CDS auction, see L. Pollack, "Mechanics," *Financial Times*, January 11, 2012.

⁵*Notional value* refers to the total face value of bonds covered by CDS contracts. The *present value* of a CDS contract at its creation is usually zero. That is, the buyer of credit protection usually pays no money up front. Then the present value fluctuates, increasing as and if credit risk increases, but is always smaller than the notional value unless the bond turns out to be totally worthless. Data on credit derivatives is published by the International Swap Dealers Association (ISDA) at www.isda.org.

⁶Governments cannot print the currencies of other countries, and may be forced into default on their foreign currency debt. For example, in 2001 Argentina defaulted on \$95 billion of foreign currency debt, and in 2012 investors in Greek government debt had to accept a write-down of about \$100 billion. Governments have occasionally defaulted on their own currency's debt. For example, in 1998, the Russian government defaulted on \$36 billion of ruble debt.

⁷Notice that the effect of an interest rate rise on the value of a put option is the opposite of its effect on the value of a call. Circular's option to extinguish the debt is more valuable when interest rates are high. However, the effect on put values of changes in the interest rate is generally modest.

⁸S. M. Schaefer and P. Feldhutter, "The Credit Spread Puzzle—Myth or Reality?" *Review of Financial Studies* 31 (2018), pp. 2897–2942.

⁹There are limits to pension payments made by the PBGC to retired employees. Employees with large pensions are not made whole.

¹⁰The pension guarantee is not ironclad. If the PBGC cannot meet its obligations, the government is not committed to providing the extra cash. But few doubt that it would do so.

¹¹W. Kiska, D. Lucas, and M. Phaup, "The Risk Exposure of the Pension Benefit Guaranty Corporation" (Washington, D.C.:

Congressional Budget Office, September 2005).

¹²The SEC has been concerned about the power wielded by the three bond-rating agencies. Therefore, it has approved seven new, nationally recognized statistical rating organizations (NRSROs): DBRS, A.M. Best, Egan-Jones Ratings, Morningstar Credit Ratings (previously known as Realpoint), Kroll Bond Rating, HR Ratings de Mexico, and Japan Credit Rating.

¹³Rating services also provide a finer breakdown. Thus, a bond might be rated A-1, A-2, or A-3 (the lowest A rating). In addition, the rating service may announce that it has put an issue on its watch list for a possible upgrade or downgrade.

¹⁴Investment-grade bonds can usually be entered at face value on the books of banks and life insurance companies.

¹⁵R. Cantor and F. Packer, “Determinants and Impact of Sovereign Credit Ratings,” Federal Reserve Bank of New York, *Economic Policy Review* 2 (October 1996), pp. 37–53.

¹⁶The most commonly used consumer credit score is the FICO score, which is used by the three main credit agencies—Experian, TransUnion, and Equifax. The agencies also use their own proprietary scoring system, VantageScore.

¹⁷For two examples that make use of both accounting and market data, see T. Shumway, “Forecasting Bankruptcy More Accurately: A Simple Hazard Model,” *Journal of Business* 74 (2001), pp. 101–124; and J. Y. Campbell, J. Hilscher, and J. Szilagyi, “In Search of Distress Risk,” *Journal of Finance* 63 (December 2008), pp. 2899–2939.

¹⁸E. Altman, “Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy,” *Journal of Finance* 23 (September 1968), pp. 589–609.

¹⁹See R. C. Merton, “On the Pricing of Corporate Debt: The Risk Structure of Interest Rates,” *Journal of Finance* 29 (1972), pp. 449–470.

²⁰If you wish to estimate the actual, rather than risk-neutral, probability that Upsilon will default, you need to use the expected return on the assets rather than the risk-free interest rate.

²¹Merton proposed an ingenious way to back out asset value and volatility from the value and volatility of the equity.

²²In Moody’s model, the probability of default is called the *expected default frequency (EDF)*.